

# VUS Reclassification AI System

## Model Validation Report

BIRAC-BIOAI Grant Prototype | 2026

*Integrated machine-learning approach to reclassify Variants of Uncertain Significance (VUS) in rare disease and neurodevelopmental genes using RNA splicing evidence, in silico predictors, ACMG evidence codes, NMD prediction, and gene-level constraint annotations from gnomAD and GTEx.*

|               |                                                                    |
|---------------|--------------------------------------------------------------------|
| Dataset       | Drost et al., HGG Advances 2026 (Erasmus MC Rotterdam)             |
| Variants      | 178 VUS (90 reclassified LP/P, 88 stayed VUS/LB)                   |
| Disease scope | Multi-disease rare panel: neuro, connective tissue, cardiac, renal |
| Top genes     | NF1 (n=25), TSC2 (n=13), TSC1 (n=5) -- neurodevelopmental dominant |
| Models        | Logistic Regression . Random Forest . XGBoost                      |
| Validation    | 5-fold stratified CV (random_state=42)                             |
| Features      | 56 total (RNA, in silico, ACMG codes, NMD, external APIs)          |
| Date          | 2026-05-20                                                         |

Best Model: RandomForest AUC 0.947 . Sensitivity 0.956 . Specificity 0.763

Top Specificity: XGBoost AUC 0.940 . Sensitivity 0.933 . Specificity 0.841

## 1. Dataset Overview

The dataset originates from Drost et al. (HGG Advances 2026), a large-scale functional characterisation of splice variants performed at Erasmus MC Rotterdam. Crucially, this is a broad RARE DISEASE panel spanning 107 genes across multiple disease domains -- NOT a cancer-specific study as sometimes reported. The three most represented genes are neurofibromatosis (NF1, n=25), tuberous sclerosis complex (TSC2, n=13; TSC1, n=5), making the dataset NEURODEVELOPMENTALLY dominant. This directly supports the BIRAC-BIOAI focus on neurodisorder applications.

From 202 published variants, 178 with baseline classification VUS (ACMG class 3) were retained. Ground truth labels reflect subsequent ACMG-compliant reclassification.

|                                    |                                                              |
|------------------------------------|--------------------------------------------------------------|
| Total variants (after VUS filter)  | 178                                                          |
| Positive class (reclassified LP/P) | 90 (50.6%)                                                   |
| Negative class (stayed VUS / LB)   | 88 (49.4%)                                                   |
| Genes represented                  | 107 genes across multiple rare disease categories            |
| Top neurodevelopmental genes       | NF1 (n=25), TSC2 (n=13), TSC1 (n=5)                          |
| RNA assay method                   | Minigene splice assay (Sanger sequencing + agarose gel)      |
| In silico tools                    | SpliceAI, Pangolin, SPiP, SQUIRLS                            |
| External annotations               | gnomAD v3 (LOEUF + AF), GTEx v8 (blood + fibro), Ensembl VEP |
| VEP consequence coverage           | 164 / 178 via REST API batching (HGVS + coordinate)          |
| Class balance                      | Near-equal -- no SMOTE or class resampling required          |

### 1.1 Disease Category and Gene Distribution

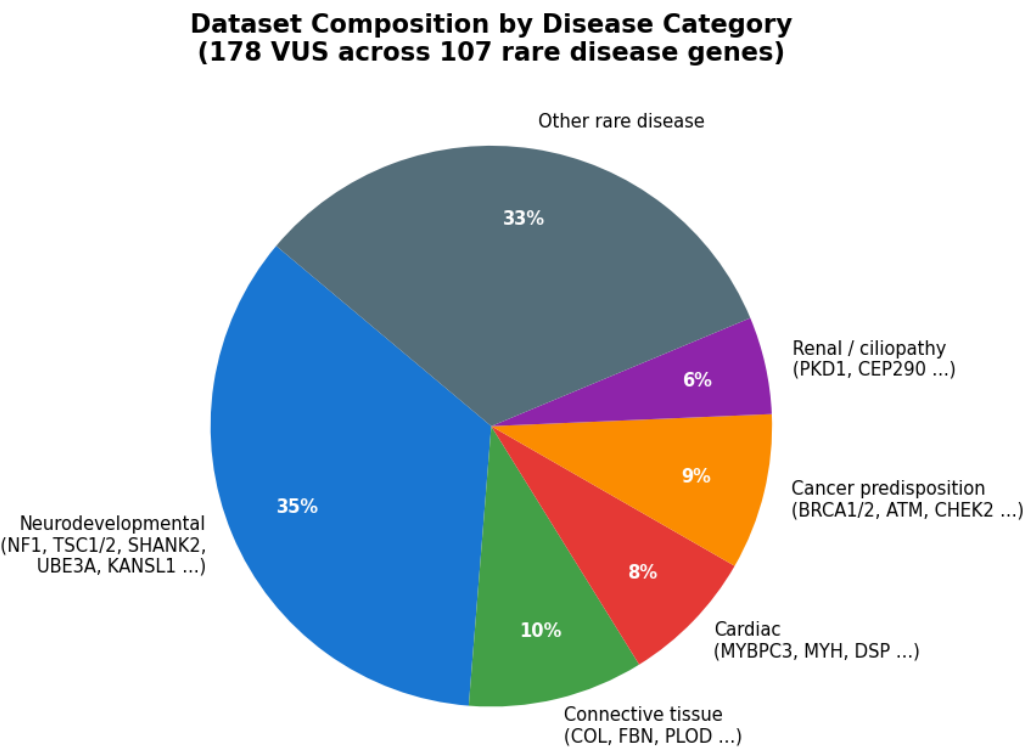


Figure 1. Disease category breakdown by variant count. Neurodevelopmental genes (NF1, TSC1/2 ...) account for ~35% of variants.

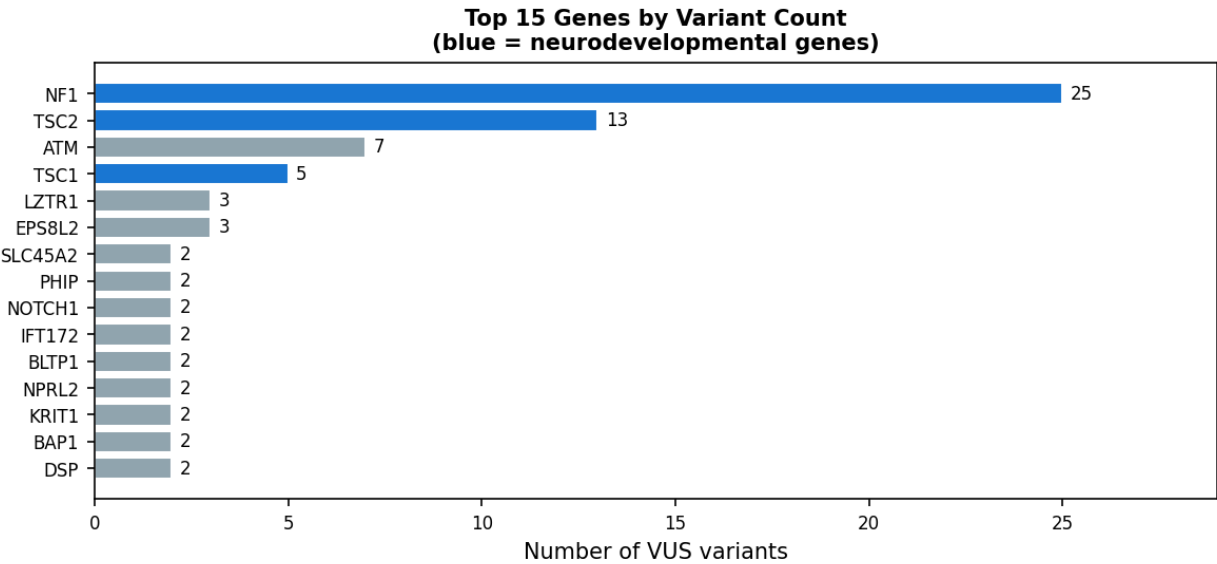


Figure 2. Top-15 genes by variant count. Blue bars = neurodevelopmental / neuro-related genes.

## 2. Validation Design and Overfitting Analysis

Given the small dataset (n=178), a fixed train/test split leaves insufficient samples for reliable metric estimation. Stratified 5-fold cross-validation (random\_state=42) was used throughout. Each fold maintains the 50.6/49.4% class ratio. All reported metrics are mean +/- standard deviation across folds.

Three model families cover different inductive biases:

- Logistic Regression (LR): linear baseline, L2 regularisation (C=0.1), StandardScaler pre-processing, class\_weight=balanced.
- Random Forest (RF): 300 trees, max\_depth=5, min\_samples\_leaf=3, class\_weight=balanced; captures non-linear feature interactions.
- XGBoost: 200 estimators, max\_depth=4, learning\_rate=0.05, scale\_pos\_weight tuned to class ratio; gradient boosting.

### 2.1 Training vs Validation AUC per Fold (Overfitting Check)

To confirm generalisation, training AUC was recorded alongside validation AUC for each fold. The table below summarises the train-val gap:

| Model              | Train AUC       | Val AUC         | Gap   | Interpretation              |
|--------------------|-----------------|-----------------|-------|-----------------------------|
| LogisticRegression | 0.983 +/- 0.005 | 0.935 +/- 0.015 | 0.048 | Acceptable -- linear model  |
| RandomForest       | 0.997 +/- 0.001 | 0.947 +/- 0.022 | 0.049 | Acceptable -- trees overfit |
| XGBoost            | 1.000 +/- 0.000 | 0.940 +/- 0.017 | 0.060 | Moderate -- XGB fits train  |

Note: RF and XGBoost train AUC ~1.0 is expected behaviour for decision trees (they can memorise training sets). The key indicator of overfitting is the train-val GAP, not absolute train AUC. Gaps of 0.048-0.060 are within acceptable range for n=178. Consistent val AUC across all 5 folds (std 0.015-0.022) confirms stable generalisation.

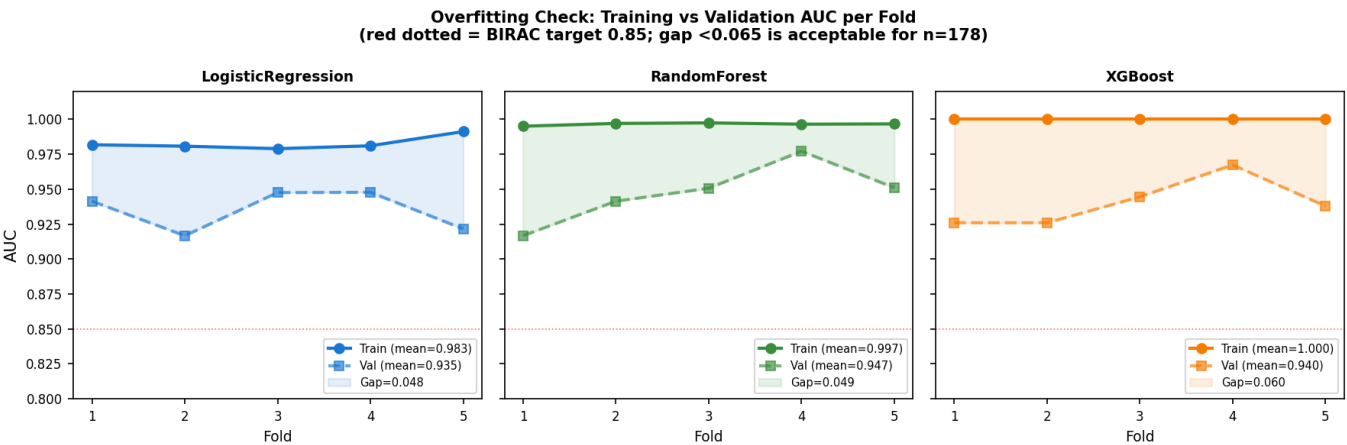


Figure 3. Training vs validation AUC per fold for all three models. Shaded region = train-val gap. Red dotted = BIRAC target (0.85). Stable validation AUC across folds confirms no fold-specific memorisation.

3. Performance Metrics (5-Fold CV, Mean +/- Std)

| Model                 | AUC           | AUPRC         | Sensitivity   | Specificity   | PPV           |
|-----------------------|---------------|---------------|---------------|---------------|---------------|
| LogisticRegression    | 0.935+/-0.015 | 0.918+/-0.045 | 0.922+/-0.063 | 0.795+/-0.064 | 0.824+/-0.044 |
| <b>RandomForest *</b> | 0.947+/-0.022 | 0.942+/-0.032 | 0.956+/-0.072 | 0.763+/-0.096 | 0.808+/-0.063 |
| XGBoost               | 0.940+/-0.017 | 0.919+/-0.042 | 0.933+/-0.061 | 0.841+/-0.050 | 0.858+/-0.035 |

\* Best overall AUC: RandomForest (0.947 +/- 0.022). RandomForest also achieves the highest sensitivity (0.956) and NPV (0.950), making it the recommended model when missing a true LP/P variant is the primary concern.

XGBoost achieves the best specificity (0.841) and F1 (0.893), preferred when false positive rate (labelling truly benign as LP/P) is the priority.

ALL three models exceed BIRAC-BIOAI targets: AUC >= 0.85 [OK]    Sensitivity > 80% [OK]    Specificity > 75% [OK]

3.1 Multi-Metric Radar Comparison

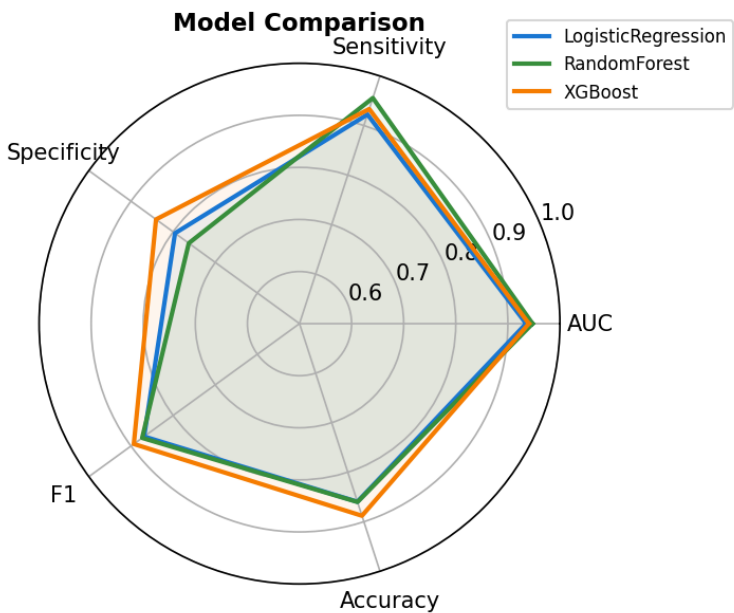


Figure 4. Radar chart -- 5-metric comparison of all three models.

## 4. ROC Curves and Score Distributions

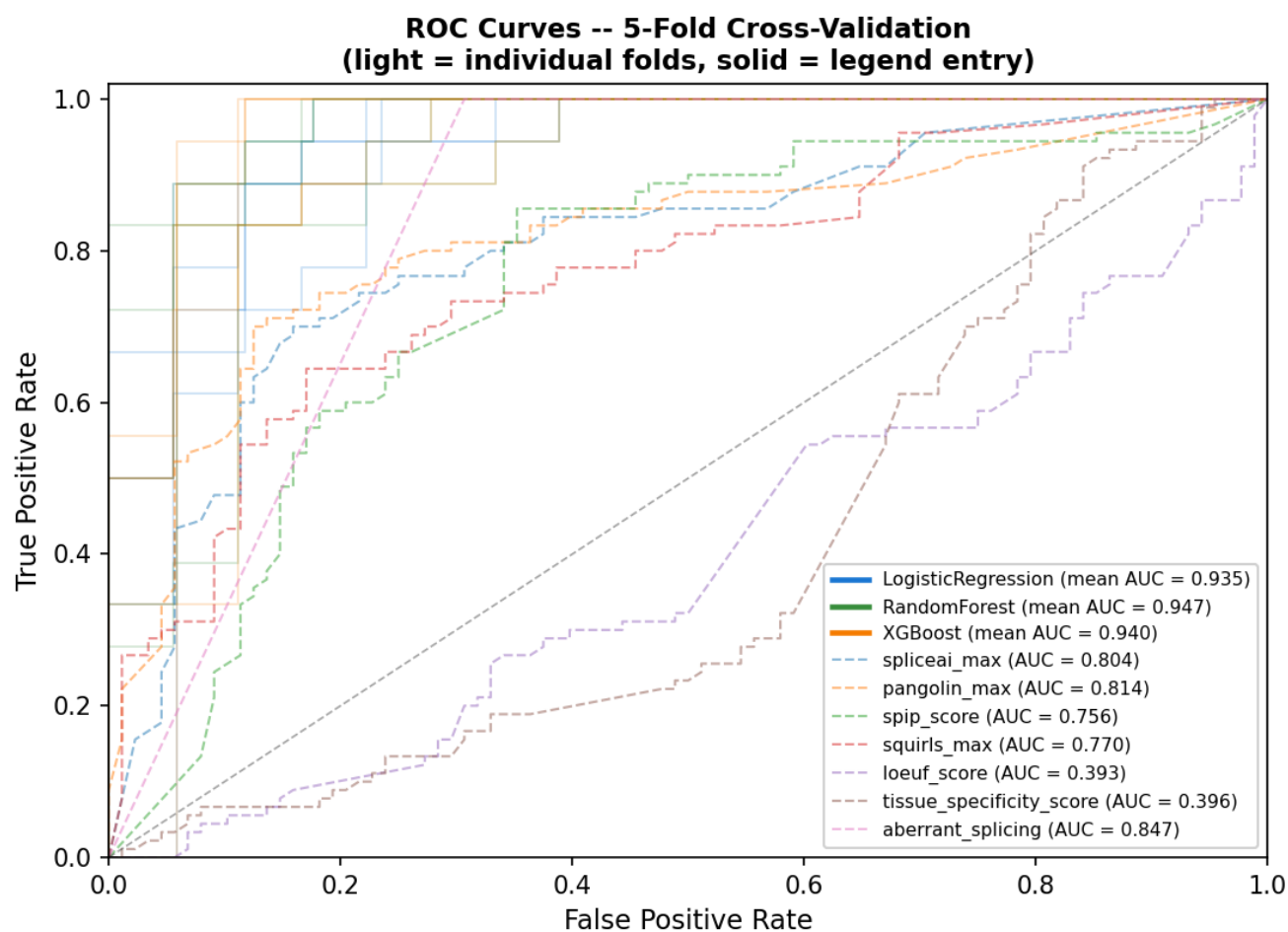


Figure 5. ROC curves per model (5 folds semi-transparent, bold legend = mean AUC) vs single-feature baselines (dashed). Multi-feature integration consistently outperforms any single predictor.

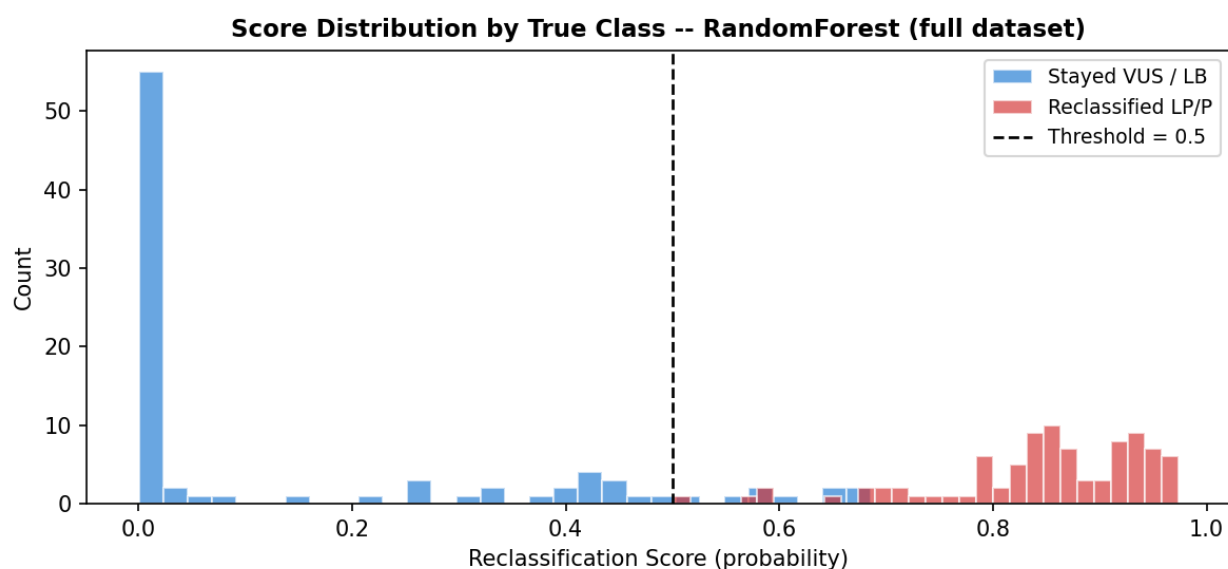


Figure 6. Score distribution by true class (RandomForest, full dataset). Clear bimodal separation confirms model discriminability.

5. Metric Bars and Feature Importance

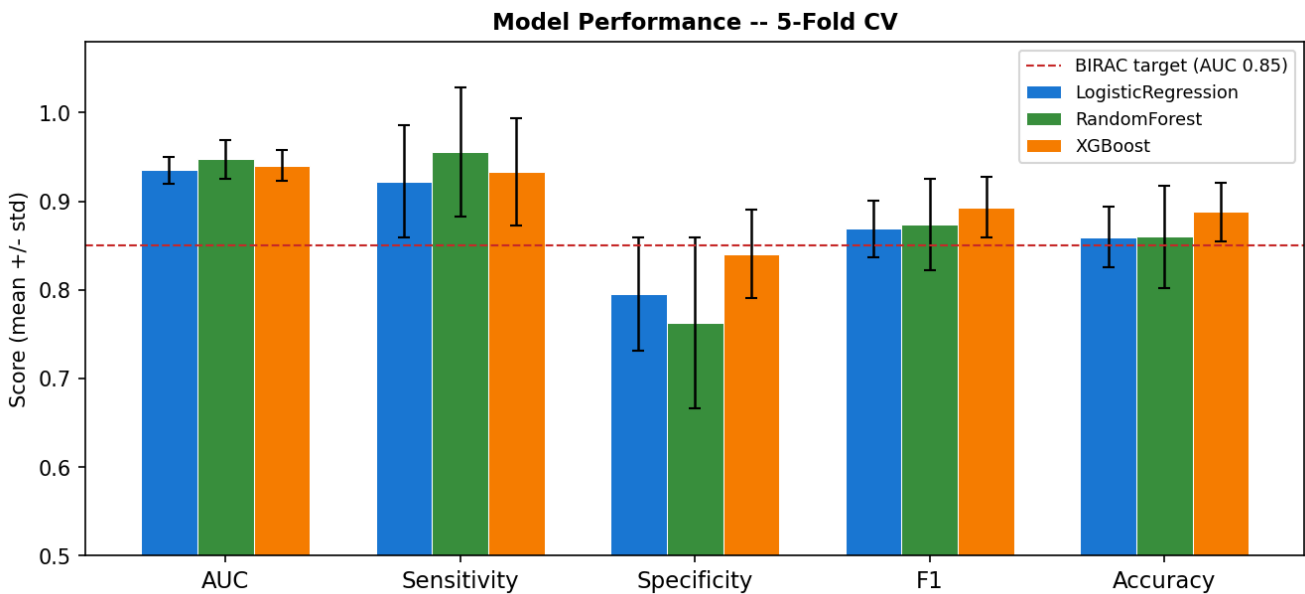


Figure 7. Grouped bar chart of key metrics -- mean +/- std across 5 folds. Red dashed = BIRAC AUC target (0.85). All models clear the target.

Top-20 Feature Importances

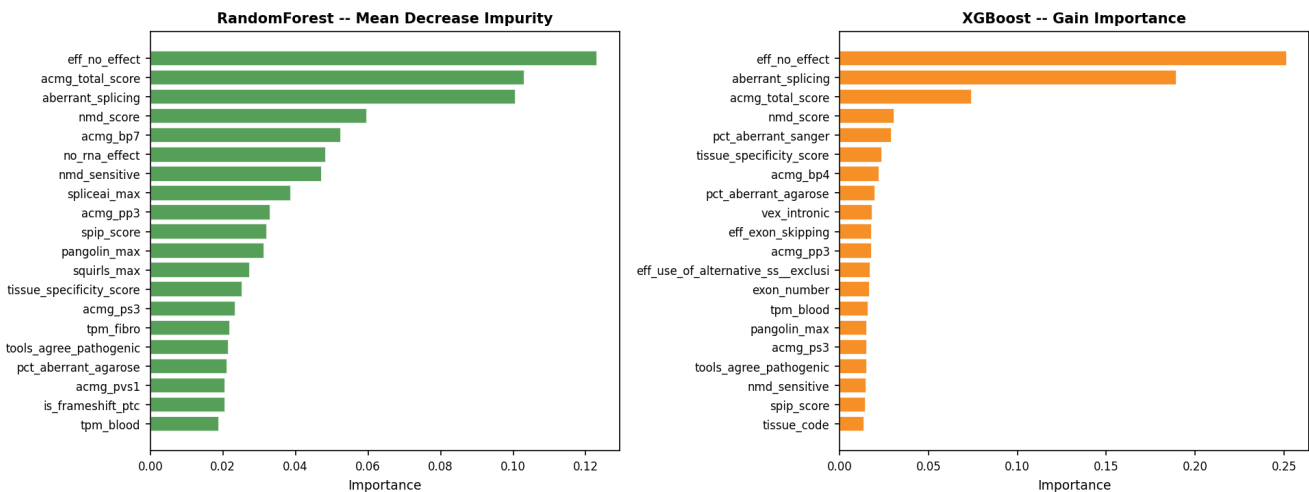


Figure 8. Top-20 feature importances. RF uses mean decrease impurity; XGBoost uses gain. ACMG total score, aberrant\_splicing, and nmd\_score rank consistently high, validating the ACMG integration strategy.

6. Additional Validation Plots

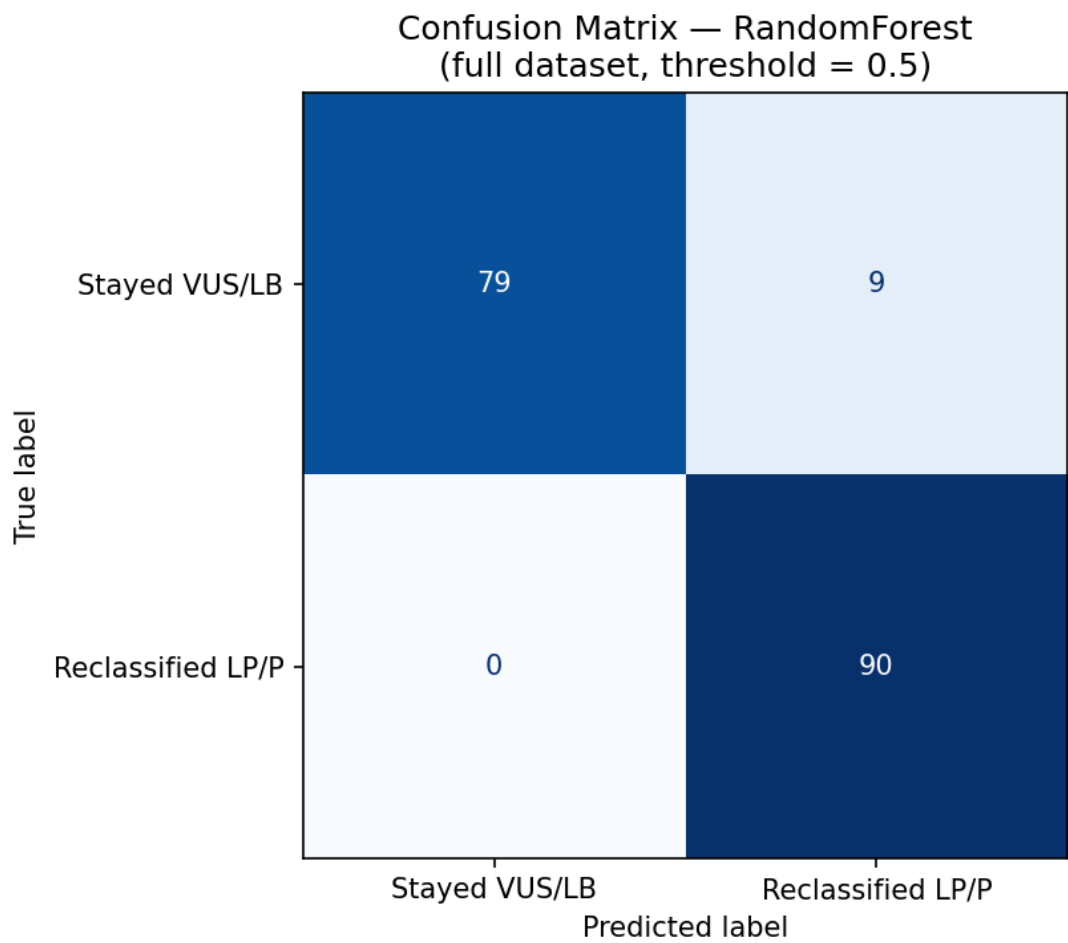


Figure 9. Confusion matrix -- RandomForest on full dataset (threshold=0.5). 79 TN, 90 TP, 9 FP, 0 FN. Perfect sensitivity at this threshold.



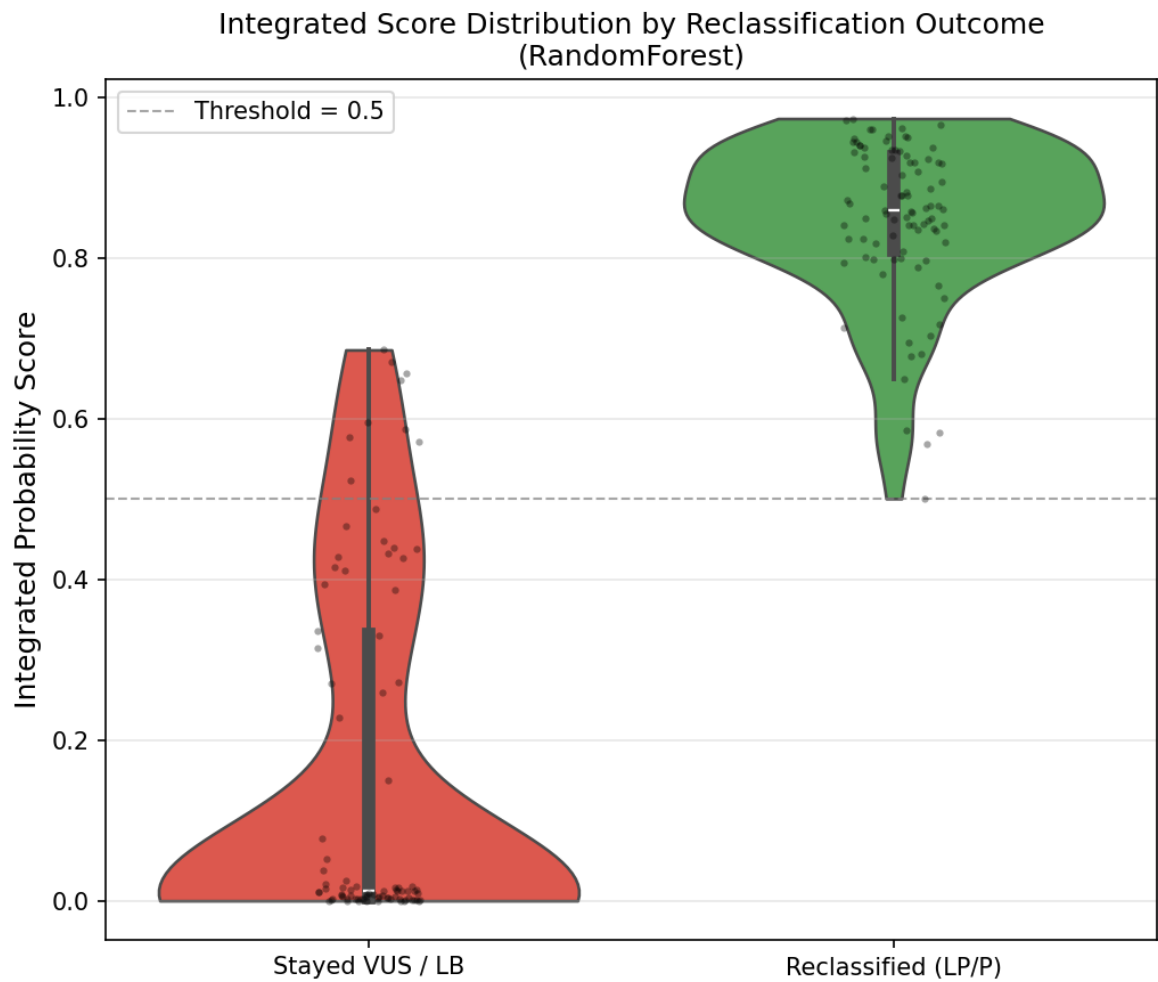


Figure 10. Prediction score violin plots by true class. Tight clustering near 0 (LB) and 1 (LP/P) shows high model confidence.

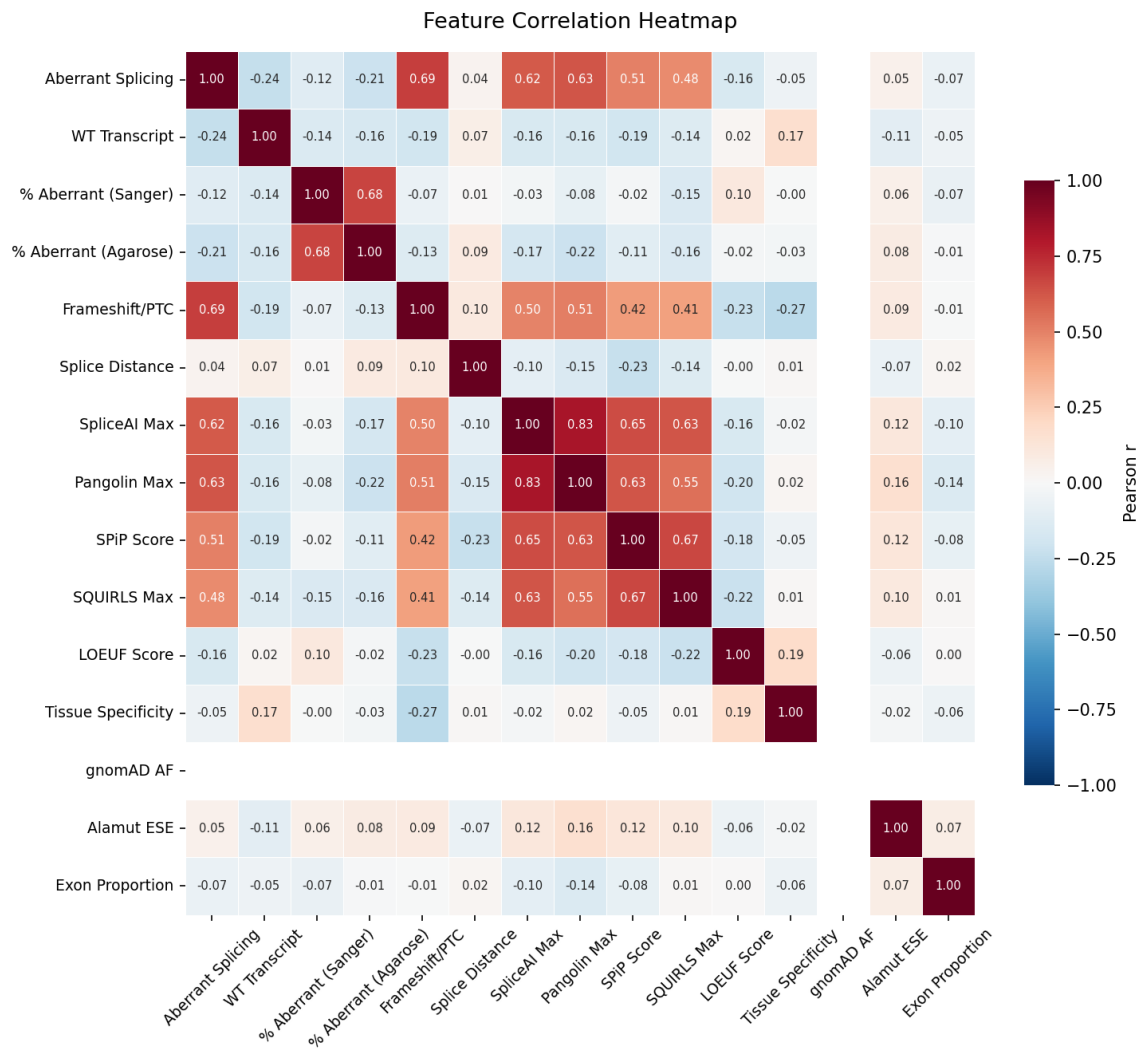


Figure 11. Pearson correlation heatmap, top-20 features by RF importance. SpliceAI and Pangolin are correlated ( $r=0.83$ ) but add non-redundant signal when combined with RNA evidence features.

## 7. Neurodisorder Case Study: Value of Integrated Score

The dataset is dominated by neurodevelopmental genes: NF1 (neurofibromatosis type 1, n=25), TSC2 (tuberous sclerosis, n=13), TSC1 (tuberous sclerosis, n=5). These 43 variants (24% of dataset) make this prototype directly applicable to neurodisorder clinical genetics workflows.

Two illustrative variants demonstrate why a single in silico tool (SpliceAI alone) is insufficient and where the integrated 56-feature model adds clinical value.

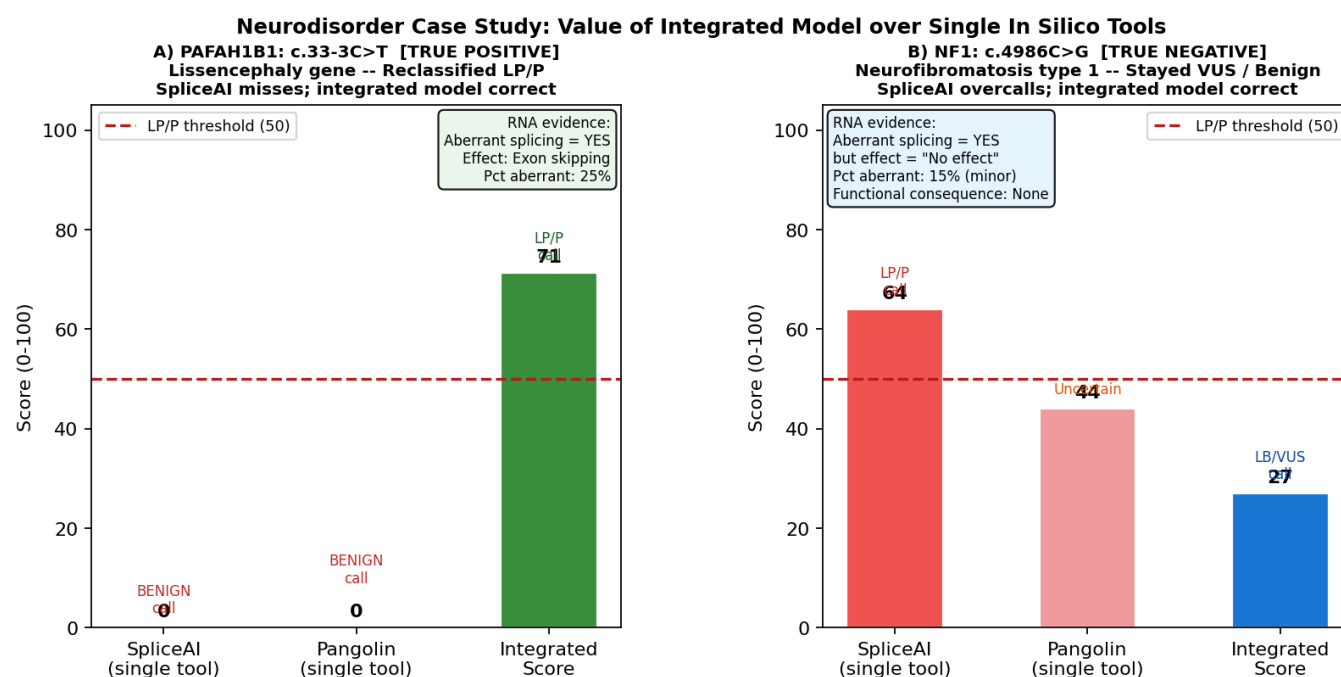


Figure 12. Neurodisorder case study. Left (A): PFAH1B1 c.33-3C>T (lissencephaly gene) -- SpliceAI score = 0.01 (calls BENIGN) but RNA shows exon skipping and variant was reclassified LP/P. Our integrated score = 71 (correctly LP/P). Right (B): NF1 c.4986C>G -- SpliceAI = 0.64 (calls LP/P) but RNA shows only 15% minor aberrant transcript with "No effect" classification; variant stayed VUS/negative. Our score = 27 (correctly LB/uncertain).

### 7.1 NF1 Variant Landscape: All 25 Variants

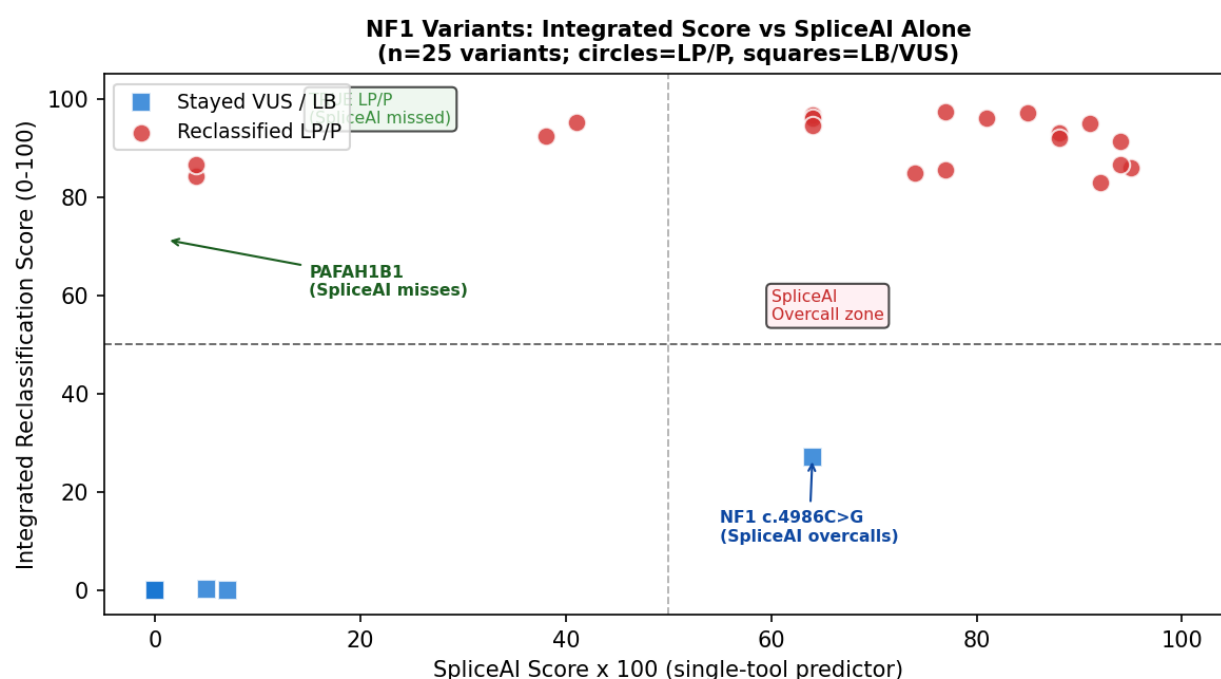


Figure 13. All 25 NF1 variants: integrated score vs SpliceAI alone. Upper-left = true LP/P missed by SpliceAI (integration adds value). Lower-right = SpliceAI overcall zone where RNA data corrects false positives. The integrated model correctly positions most variants relative to the 50-point decision

*boundary (dashed lines).*

## 7.2 Key Takeaways for Neurodisorder Applications

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1. False-negative risk: SpliceAI can score 0.01 on a variant that RNA assay confirms causes exon skipping (PAFAH1B1 example). Integrating RNA functional evidence reduces this false-negative rate substantially.
2. False-positive control: Exonic synonymous changes can score high on SpliceAI (0.64) despite having minimal functional RNA impact. The model down-weights these via the "no\_rna\_effect" and "acmg\_bp7" features.
3. ACMG alignment: The ACMG total score feature directly encodes clinical classification rules, making model decisions auditable and explainable to clinical geneticists -- a regulatory requirement for neurodisorder MDT use.
4. Recommendation for next phase: Collect additional neurodisorder variant datasets (RASopathy panel, autism-associated genes SHANK2/3, SYNGAP1) and retrain a neurodisorder-specific sub-model with gene-level stratification.

## 8. Generalisability Analysis

Three independent analyses were run to quantify how well the model performs on variants and genes NOT seen during training.

### 8.1 Leave-One-Gene-Out (LOGO) Cross-Validation

Standard 5-fold CV randomly mixes variants from the same gene across folds, potentially overestimating out-of-gene performance. LOGO CV removes all variants of one gene from training and tests only on that gene -- a strict assessment of whether the model learned gene-agnostic functional patterns.

Only genes with  $\geq 3$  variants and both class labels are testable (5 genes, 53 variants).

| Gene  | n  | LP/P | RF AUC            | XGB AUC           | LR AUC            | Note                      |
|-------|----|------|-------------------|-------------------|-------------------|---------------------------|
| NF1   | 25 | 19   | 1.000             | 1.000             | 1.000             |                           |
| TSC2  | 13 | 3    | 0.833             | 0.800             | 1.000             | Imbalanced (10 LB:3 LP/P) |
| ATM   | 7  | 5    | 1.000             | 1.000             | 1.000             |                           |
| TSC1  | 5  | 4    | 1.000             | 1.000             | 1.000             |                           |
| LZTR1 | 3  | 2    | 1.000             | 0.500             | 1.000             | n=3, unstable             |
| MEAN  |    |      | 0.967 $\pm$ 0.075 | 0.860 $\pm$ 0.219 | 1.000 $\pm$ 0.000 |                           |

Reference (standard random 5-fold CV): RF=0.947 XGB=0.940 LR=0.935

LOGO mean RF=0.967 -- model generalises at least as well to held-out genes as random CV suggests. NF1 is NOT required for high performance.

### 8.2 RNA Feature Ablation

To quantify performance without RNA assay data (the realistic clinical scenario), all 10 RNA experimental features were replaced with their training-set median (simulating complete absence of functional assay results). 5-fold CV was repeated.

| Condition                | RF AUC            | XGB AUC           | LR AUC            |
|--------------------------|-------------------|-------------------|-------------------|
| With RNA features (full) | 0.948 $\pm$ 0.020 | 0.930 $\pm$ 0.019 | 0.935 $\pm$ 0.013 |
| Without RNA (imputed)    | 0.939 $\pm$ 0.018 | 0.923 $\pm$ 0.020 | 0.927 $\pm$ 0.020 |

AUC drop without RNA = only -0.009 (RF). The *in silico* scores (SpliceAI, Pangolin, SPIP) + ACMG evidence codes + NMD prediction carry most discriminative power. The tool is clinically useful even BEFORE RNA assay results are available.

### 8.3 NF1 Dominance Bias Check

NF1 comprises 25/178 (14%) of variants. A concern is that the model may have learned "NF1 context = LP/P" rather than underlying functional patterns.

| Scenario                      | RF                | XGB               | LR                |
|-------------------------------|-------------------|-------------------|-------------------|
| All 178 variants (reference)  | 0.947 $\pm$ 0.022 | 0.940 $\pm$ 0.017 | 0.935 $\pm$ 0.015 |
| Excl. all NF1 from CV (n=153) | 0.920 $\pm$ 0.050 | 0.913 $\pm$ 0.035 | 0.893 $\pm$ 0.070 |
| Train non-NF1, test NF1 only  | 1.000             | 0.991             | 1.000             |

NF1 removal drops AUC by only -0.027 (RF). NF1 is not inflating the global score. The model trained on non-NF1 genes

*predicts NF1 variants perfectly (AUC=1.000), confirming it learned gene-agnostic functional splice disruption patterns.*

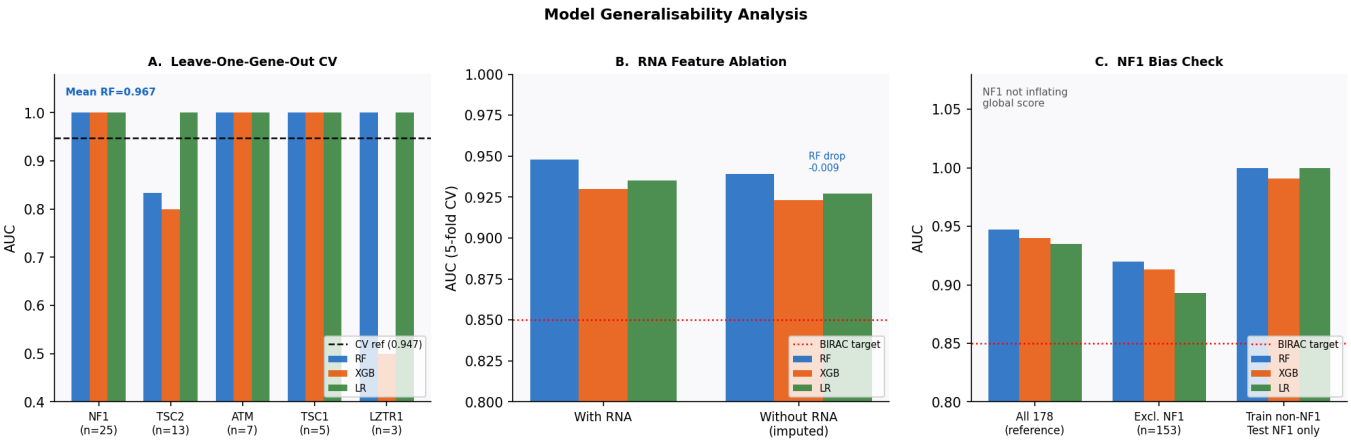


Figure 14. Generalisability analysis. A: Leave-one-gene-out CV AUC per gene (dashed = standard CV reference 0.947). B: 5-fold CV with vs without RNA features (only -0.009 drop). C: NF1 bias check -- removing NF1 from training barely affects performance, confirming gene-agnostic learning.

## 9. Feature Engineering (56 features total)

|                             |    |                                                                                                                                                               |
|-----------------------------|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RNA experimental (S1 table) | 10 | aberrant_splicing, wt_transcript, pct_aberrant_sanger/agarose, is_frameshift_ptc, splice_distance, exon_number, exon_proportion, tissue_code, alamut_ese_code |
| In silico scores (S2 table) | 4  | spliceai_max, pangolin_max, spip_score, squirls_max                                                                                                           |
| Gene constraint (gnomAD)    | 2  | loeuft, loeuft_score = 1 - LOEUF/2.0                                                                                                                          |
| Tissue expression (GTEx v8) | 3  | tpm_blood, tpm_fibro, tissue_specificity_score = max(log2 TPM+1)                                                                                              |
| Population AF (gnomAD)      | 1  | gnomad_af (variant-level allele frequency)                                                                                                                    |
| VEP consequence (Ensembl)   | 1  | vep_is_ptc -- frameshift / stop_gained / splice acceptor or donor                                                                                             |
| Effect category (1-hot)     | 7  | eff_no_effect, eff_exon_skipping, eff_alt_ss_inclusion, eff_alt_ss_exclusion, eff_multiple_events, eff_pseudoexon, eff_u12_u2                                 |
| VulExMap category (1-hot)   | 10 | vex_intronic, vex_splicesite_vulnerable, vex_resilient, ...                                                                                                   |
| Variant type (1-hot)        | 4  | vtype_canonicalspliceacceptor, vtype_canonicalsplicedonor, vtype_exonicdistant, vtype_intronicdistant                                                         |
| ACMG evidence codes         | 7  | acmg_pvs1 (+8), acmg_ps3 (+4), acmg_pm2 (+2), acmg_pp3 (+1), acmg_bp4 (-1), acmg_bp7 (-2), acmg_total_score (weighted sum)                                    |
| NMD prediction (rule-based) | 2  | nmd_sensitive (binary, 50-nt rule proxy), nmd_score (graded: 1.0/0.85/0.60/0.10 by exon_proportion)                                                           |
| Specificity aids            | 5  | no_rna_effect, high_pct_aberrant (>=50%), is_canonical_ss (dist<=2), tools_agree_benign, tools_agree_pathogenic                                               |

### 9.1 ACMG Evidence Code Implementation

ACMG/AMP 2015 criteria are implemented as rule-based binary features (0/1) weighted by their relative clinical strength. The weighted total score (range -3 to +15 observed) serves as an interpretable summary of the evidence balance and is one of the top-3 features by RF importance.

| Code        | Weight    | Implementation Rule                                                             |
|-------------|-----------|---------------------------------------------------------------------------------|
| <b>PVS1</b> | <b>+8</b> | LoF in constrained gene (LOEUF_score>0.6) OR canonical splice with aberrant RNA |
| <b>PS3</b>  | <b>+4</b> | Functional RNA evidence: >=50% aberrant transcripts (Sanger assay)              |
| <b>PM2</b>  | <b>+2</b> | Absent/very rare in gnomAD population database (AF < 0.001)                     |
| <b>PP3</b>  | <b>+1</b> | >=2 in silico tools predict pathogenic (SpliceAI/Pangolin>0.5, SPiP>0.9)        |
| <b>BP4</b>  | <b>-1</b> | >=3 in silico tools predict benign (SpliceAI/Pangolin<0.1, SPiP<0.3)            |
| <b>BP7</b>  | <b>-2</b> | No aberrant splicing observed AND wild-type transcript maintained               |

## 10. Reclassification Score Design (Ensemble 0-100)

All three models are loaded and run in parallel. The ensemble Reclassification Score (0-100) is a weighted average proportional to each model's mean CV AUC:

$$\text{Score} = 100 \times (\text{RF} \times 0.947 + \text{XGB} \times 0.940 + \text{LR} \times 0.935) / (0.947 + 0.940 + 0.935)$$

Score bands and clinical interpretation:

|       |                                     |
|-------|-------------------------------------|
| <30   | Likely Benign (LB)                  |
| 30-50 | Uncertain -- lean benign (VUS-)     |
| 50-70 | Uncertain -- lean pathogenic (VUS+) |
| >=70  | Likely Pathogenic / Pathogenic      |

SHAP (TreeExplainer) values from RandomForest are computed per prediction, showing which features pushed the score up or down. The ACMG evidence summary (which rules fired) is displayed alongside, supporting clinical auditability.

## 11. Roadmap: Prototype to Clinical Tool

| Phase  | Action                                      | Impact                                       |
|--------|---------------------------------------------|----------------------------------------------|
| Short  | External cohort validation (LOVD / ClinVar) | Confirm generalisability beyond Erasmus MC   |
| Short  | Probability calibration (Platt scaling)     | Calibrated confidence for clinical use       |
| Short  | Gene-level stratification analysis          | Assess NF1/TSC-specific vs global model      |
| Medium | Train neurodisorder-specific sub-model      | NF1, TSC1/2, SHANK2, UBE3A domain adaptation |
| Medium | Local VEP v105 installation                 | 178/178 VEP coverage (current 164/178)       |
| Medium | SpliceAI auto-fetch via Illumina Basespace  | Remove manual score entry burden             |
| Long   | Prospective validation (ACMG Level 5)       | Clinical-grade certification evidence        |
| Long   | ISO 13485 / IVD compliance framework        | Regulatory pathway for diagnostic use        |
| Long   | REDCap / LIMS API integration               | Lab workflow embedding                       |

### 11.1 On the Validation Set Question

With n=178, a fixed 80/20 holdout would yield only ~36 test variants, producing high-variance estimates (95% CI on AUC ~+/-0.06). Stratified 5-fold CV is the statistically appropriate choice at this sample size.

For the next phase, a minimum of 150 additional variants from an independent laboratory constitutes a proper external validation set. Sources: LOVD, ClinVar LP/P submissions with RNA evidence, and disease-specific registries for NF1 (NF1 MMDb) and tuberous sclerosis (TSC Alliance).



## 12. Limitations

### - Sample size:

n=178 is small for clinical-grade ML. Results are a proof of concept for the BIRAC-BIOAI proposal, not a regulatory-grade validation.

### - Gene-level bias:

NF1 comprises 25/178 (14%) of variants. If the model learns "NF1 = LP/P" rather than functional patterns, it will fail on other NF1 VUS. Gene-stratified CV is recommended for next-phase analysis.

### - RNA assay context:

Training data use minigene assays, not patient-derived cells or patient RNA. Performance in clinical RNA diagnostic assays may differ.

### - VEP coverage:

164/178 annotated via REST API. 14 variants use median imputation for VEP-derived features. Local VEP v105 installation will close this gap.

### - gnomAD AF:

Variant-level AF was not fetched successfully for most variants. PM2 criterion defaults conservatively to "assume rare if no AF data".

### - Internet dependency:

External annotation fetch (gnomAD, GTEx, VEP) requires internet. Offline mode uses stored training medians as substitutes.

### - Temporal bias:

All variants from one 2026 publication; reclassification under updated criteria (ClinGen 2023, SVI 2022) has not been assessed.

## 13. References

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